

$$y = mx + c$$

m : 變化率

m: rate of change

c : 初始值

c: initial value

x 每增加 1，y 改變多少？

How much does y change for each extra 1 in x?

當 $x = 0$ 時，y 是多少？

What is y when $x = 0$?

一條直線描述 x 與 y 的關係

A straight line describes the relationship between x and y

C : 初始值
c: the initial value

水箱原有 5 公升水

A tank starts with 5 litres of water

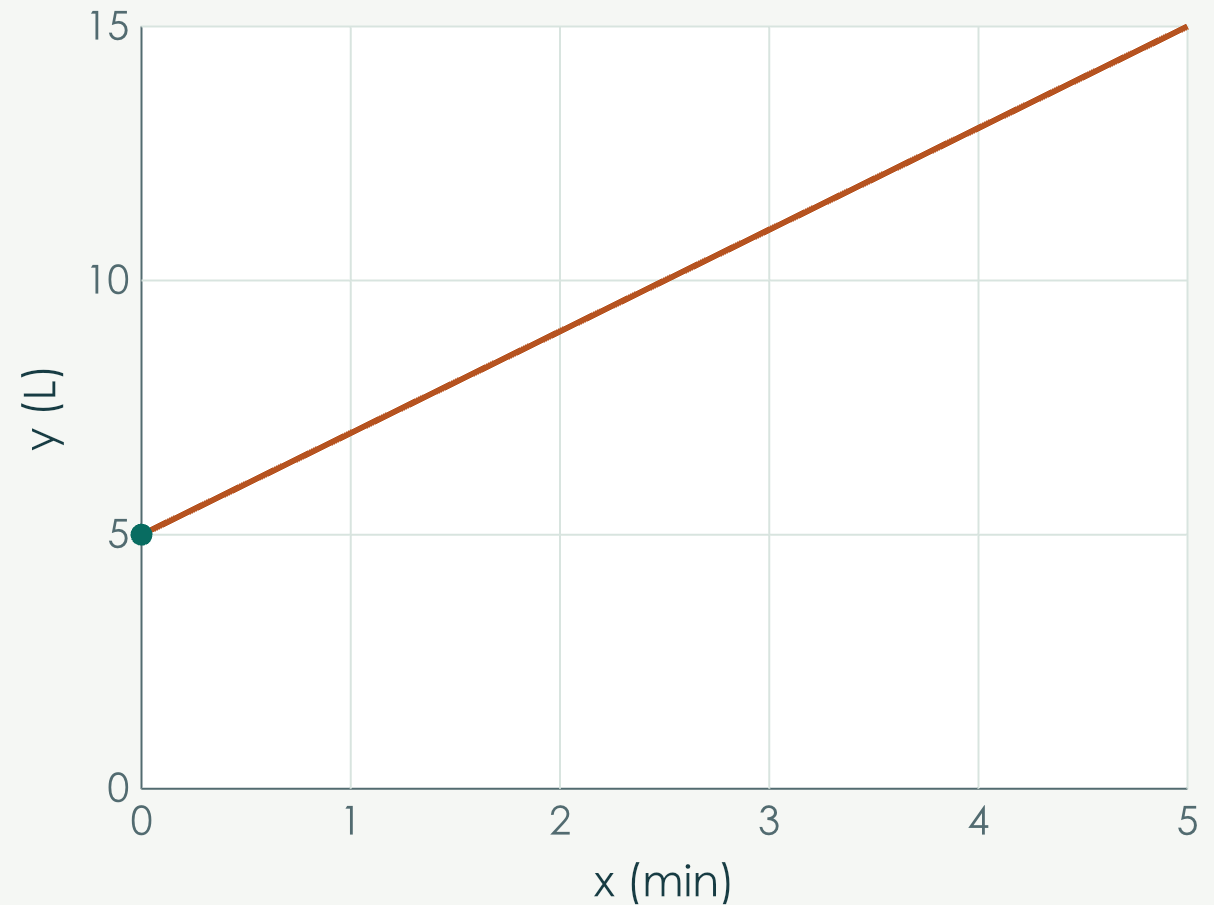
x : 時間 (分鐘)

x: time in minutes

y : 水量 (公升)

y: amount of water in litres

$x = 0, y = 5$



m : 每增加 1 單位的變化
m: the change for each extra unit

每分鐘加入 2 公升水

We add 2 litres each minute

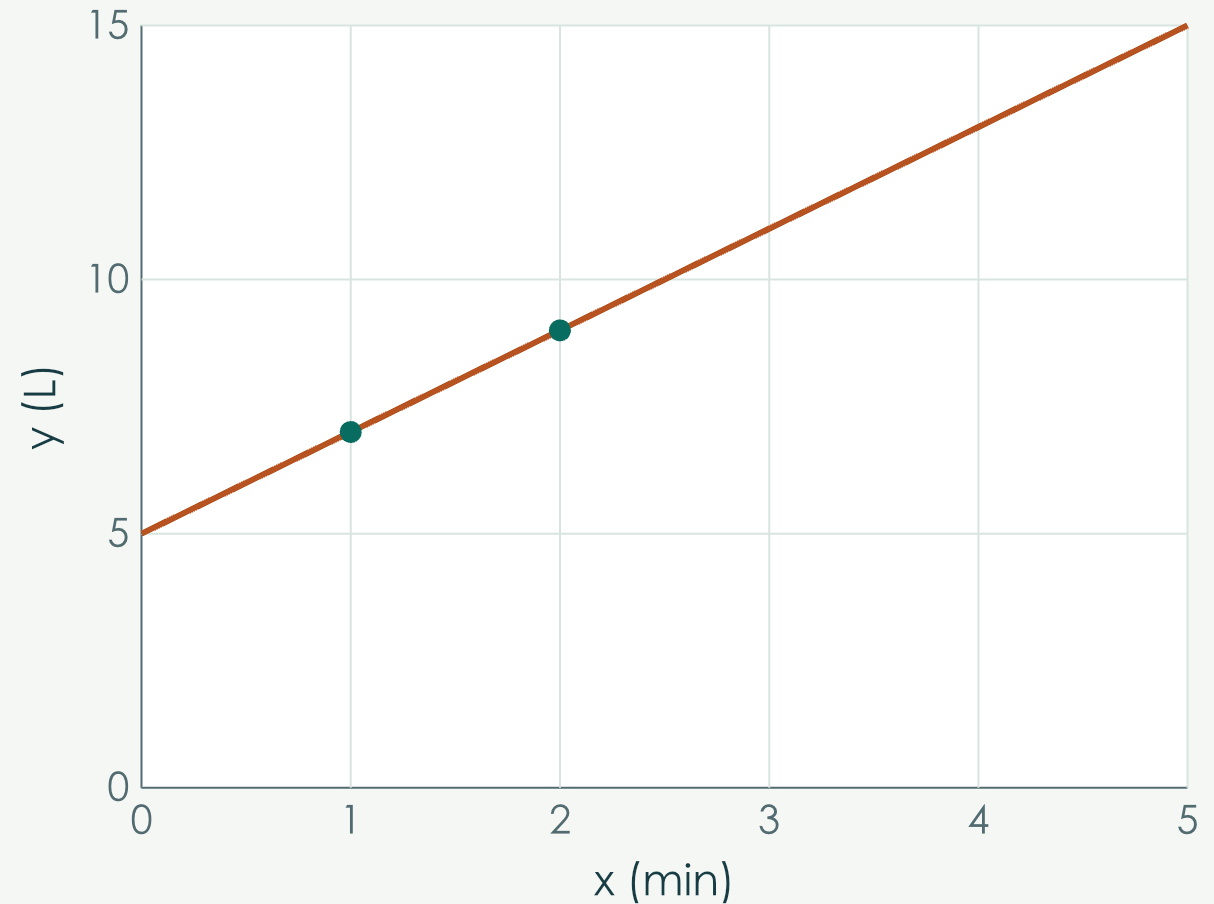
$$y = 2x + 5$$

x 增加 1 , y 增加 2

When x increases by 1, y increases by 2

m = 2 公升 / 分鐘

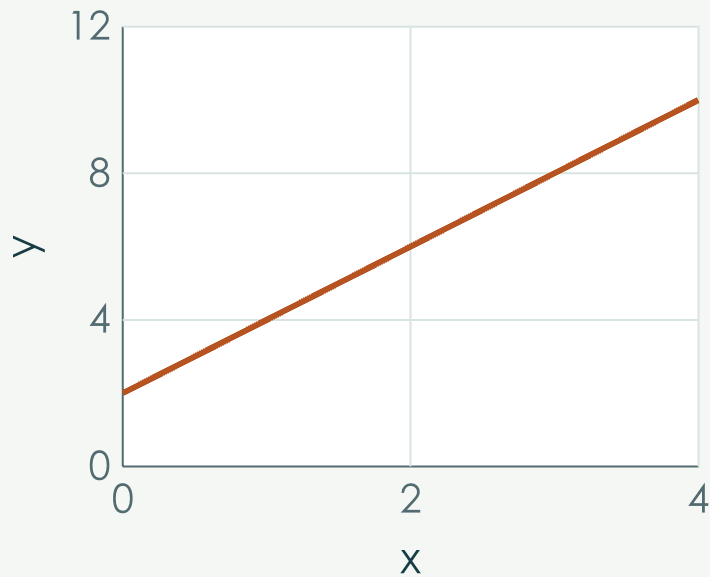
m = 2 litres per minute



m 的正負與趨勢

The sign of m and the trend

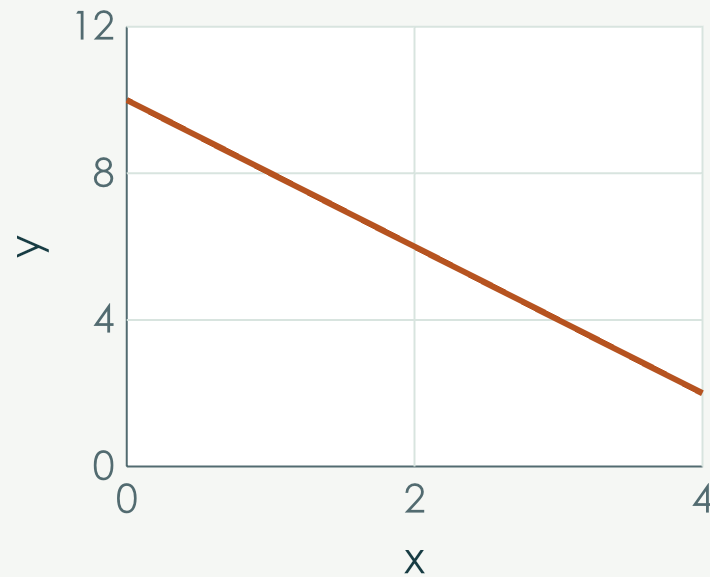
m > 0



x 增加時，y 增加

y increases as x increases

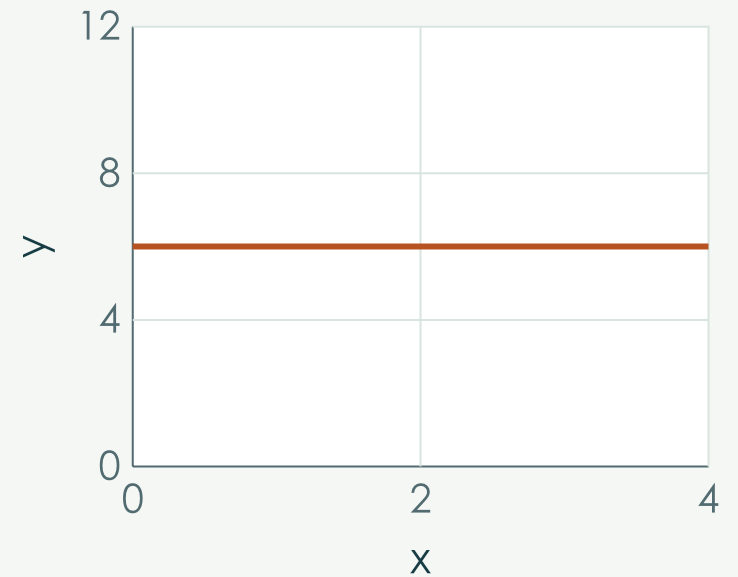
m < 0



x 增加時，y 減少

y decreases as x increases

m = 0

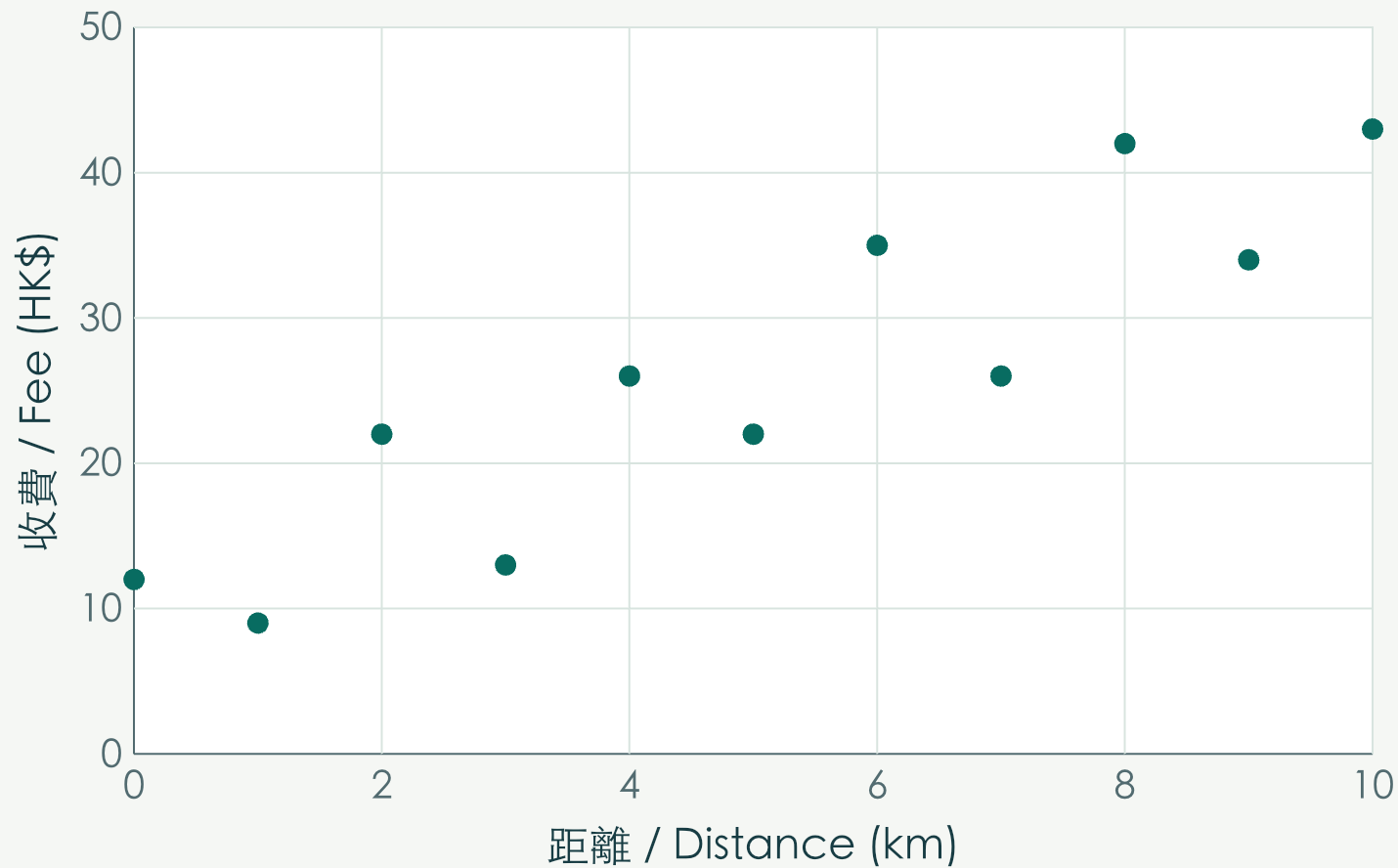


y 保持不變

y stays the same

真實情境的數據會有差異

Data in real situations varies



每個點是一個觀察

Each dot is an observation

各點是否在同一直線上？

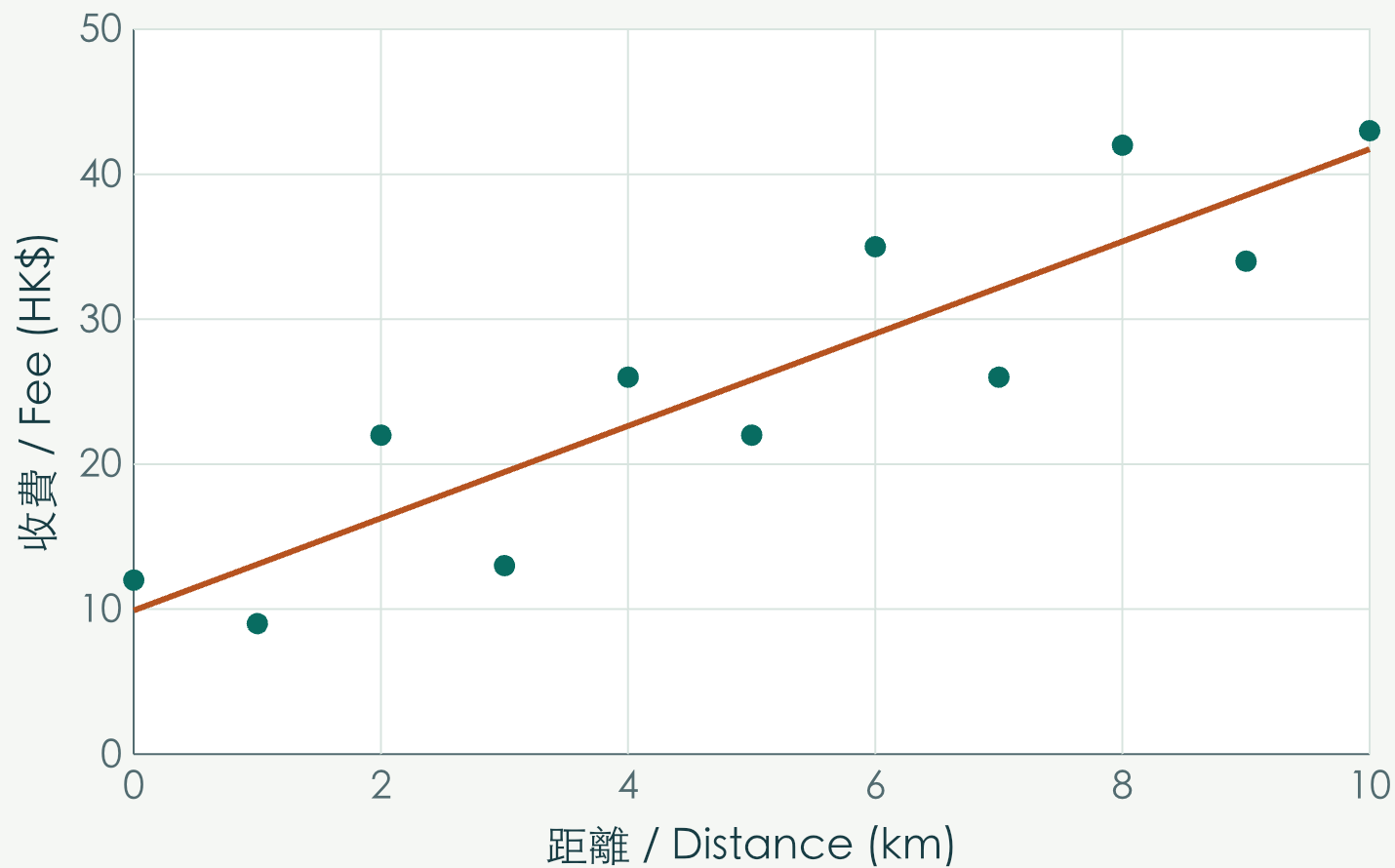
Do all the dots lie on one line?

你看到甚麼趨勢？

What trend do you notice?

線性回歸與最佳擬合直線

Linear regression and the best-fit line



用一條直線概括整體趨勢

One line summarises the overall trend

它不必通過每一個點

It need not pass through every dot

網頁會自動更新直線

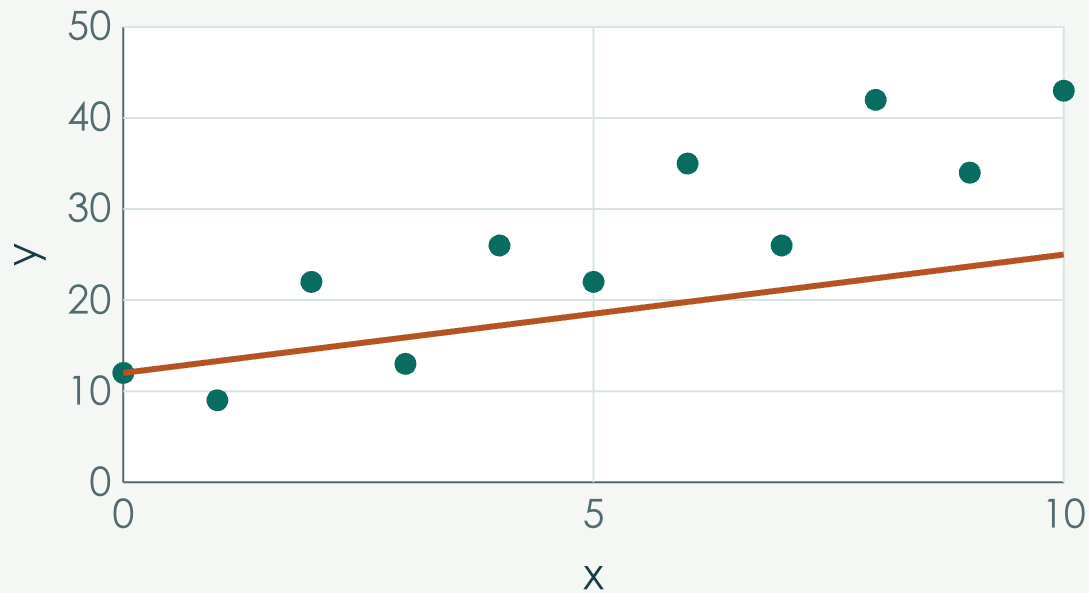
The page updates the line automatically

線性回歸是一種簡單的機器學習

Linear regression is a simple form of machine learning

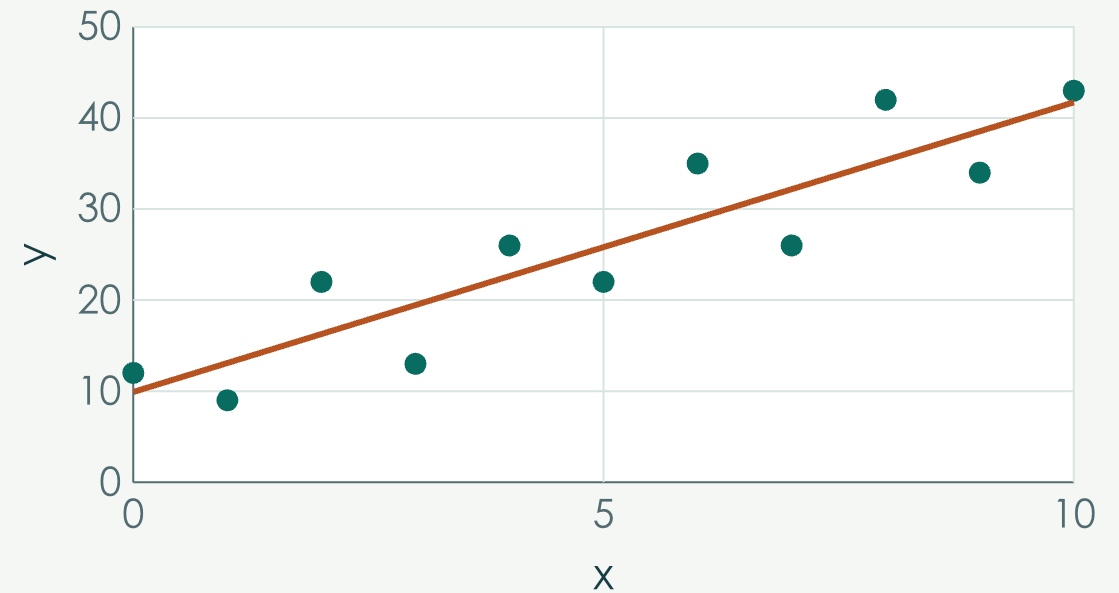
調整前的猜測（示意）

An initial guess (illustration)



根據例子，找出較合適的直線

The examples guide a better-fitting line



電腦從數據學習，然後用學到的模式作預測

The computer learns from data, then uses the pattern to predict

現代 AI 能學習更複雜的模式

Modern AI can learn more complex patterns

今天：從數字的例子，學習一條直線

Today: learn a straight line from numerical examples

許多現代 AI 使用神經網絡，學習更複雜的模式

Many modern AI systems use neural networks to learn more complex patterns

例子：辨認照片中的物件，或預測下一小段文字

Examples: recognise objects in photos, or predict the next piece of text

共同想法：從例子學習，再應用於新的情況

Shared idea: learn from examples, then apply the learning to new cases

學會模式，還是只記住例子？

Learning a pattern, or only remembering examples?

學習時：用一些例子找出模式

During learning: use some examples to find a pattern

檢查時：用新例子，比較預測與實際結果

During checking: compare predictions with actual outcomes on new examples

數據的品質會影響預測；模型的答案也可能有誤

Data quality affects predictions. A model can still give a wrong answer.

加入新點，觀察直線
Add points and observe the line

在圖上點一下，加入一個數據點
Click or tap the graph to add a data point

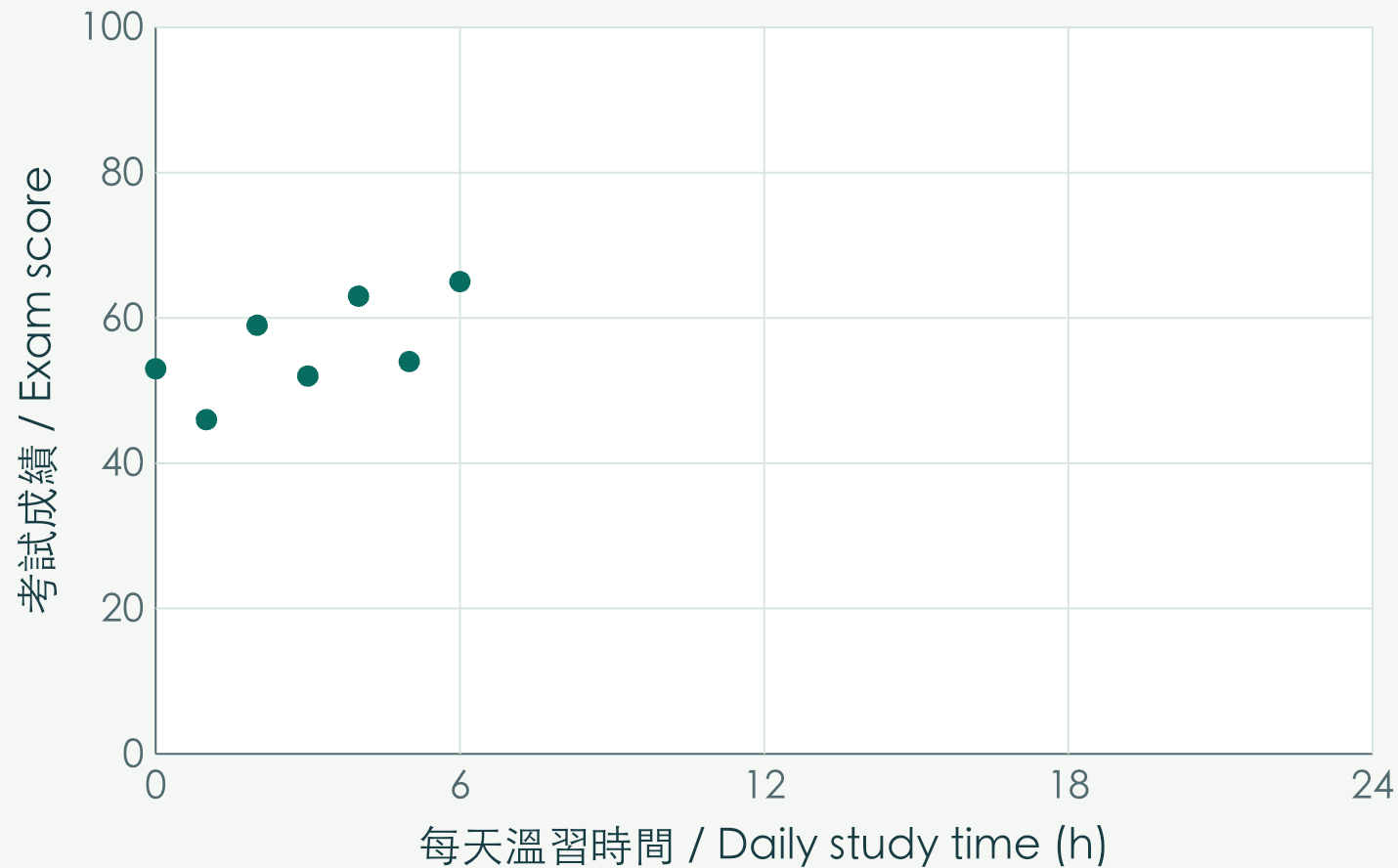
比較加入點前後的 m 和 c
Compare m and c before and after adding the point

試一個遠離其他點的新點
Try a point far from the other dots

先預測，再觀察。把結果寫在工作紙上。
Predict first, then observe. Record the result on your worksheet.

每天溫習時間與考試成績

Daily study time and exam score



3.5 小時的預測成績？

Score for 3.5 hours a day?

先畫直線，再估計

Draw your line, then estimate

10 小時的預測可信嗎？

How reliable is a prediction for 10 hours?

不同學生的模擬數據 / Simulated data for different students

預測結果與模型的限制

Prediction results and limits of the model

$$y = 2x + 50$$

$m = 2$: 模型中每小時約多 2 分

In this model, $m = 2$ marks per extra hour

$c = 50$: 0 小時時估計的成績

$c = 50$: estimated score at 0 hours

3.5 小時 : 約 57 分

3.5 hours: about 57 marks

10 小時 : 約 70 分

10 hours: about 70 marks

10 小時超出觀察範圍。這些數據也不能證明，溫習會令分數上升。

10 hours is outside the data range. The data does not prove cause and effect.

你的預測與觀察

Your prediction and observation

你的估計與網頁的預測接近嗎？

Was your estimate close to the page's prediction?

你畫的直線與網頁的直線有甚麼分別？

How does your own line compare with the line on the page?

你會較信賴哪個預測？為甚麼？

Which prediction would you trust more? Why?

工作紙最後一節 / Final section of the worksheet